Solution to Q10: Information Available in Continuous Random Variables

Problem Setup

We have (T, P) jointly continuous with independent normals:

$$T \sim N(0,1), \quad P \sim N(0,4), \quad T \perp P.$$

Joint density:

$$f_{T,P}(t,p) = \frac{1}{4\pi} \exp\left(-\frac{t^2}{2} - \frac{p^2}{8}\right), \quad (t,p) \in \mathbb{R}^2.$$

Define:

$$X(t,p) = t, \quad Y(t,p) = (t,p).$$

1. Probability space

$$\Omega = \mathbb{R}^2, \quad \mathcal{F} = \mathcal{B}(\mathbb{R}^2), \quad P(A) = \int_A f_{T,P}(t,p)\,dt\,dp.$$

2. σ-algebra generated by X

$$\sigma(X) = \{B \times \mathbb{R} : B \in \mathcal{B}(\mathbb{R})\}.$$

3. σ-algebra generated by Y

Since Y is the identity map,

$$\sigma(Y) = \mathcal{B}(\mathbb{R}^2).$$

4. Inclusion $\sigma(X) \subseteq \sigma(Y)$

Any $B \times \mathbb{R}$ is a Borel set in $\mathbb{R}^2$, so $\sigma(X) \subset \sigma(Y)$.
Y contains more information: it records both T and P, while X records only T.

5. Marginal density of X

$$f_X(t) = \frac{1}{\sqrt{2\pi}} e^{-t^2/2}.$$

6. Marginal density of Y

$$f_Y(t,p) = f_{T,P}(t,p) = \frac{1}{4\pi} \exp\left(-\frac{t^2}{2} - \frac{p^2}{8}\right).$$

7. Probabilities

$$\begin{aligned}
P(X \le 0) &= \Phi(0) = 0.5, \\
P(X > 1) &= 1 - \Phi(1) \approx 0.1587, \\
P(Y \in (-\infty, 0] \times \mathbb{R}) &= P(T \le 0) = 0.5, \\
P(Y \in [-1,1] \times [-2,2]) &= P(T \in [-1,1]) \cdot P(P \in [-2,2]) \\
&= [\Phi(1) - \Phi(-1)]^2 \approx (0.6826)^2 \approx 0.4659.
\end{aligned}$$

8. Conditional distribution of X given $Y = (t, p)$

Given $Y = (t, p)$, $X = t$ exactly:

$$P(X = t \mid Y = (t, p)) = 1.$$

In density sense, it's a Dirac delta δ_t.

9. Conditional distribution of Y given $X = t$

$X = t \Rightarrow T = t$, and P independent $N(0, 4)$:

$$f_{Y|X}(t, p) = \delta(t - t_0) \cdot \frac{1}{\sqrt{8\pi}} e^{-p^2/8}.$$

More formally: $P(P \in A \mid X = t) = \int_A \frac{1}{\sqrt{8\pi}} e^{-p^2/8} dp$.

10. Interpretation of conditioning

Conditioning on X: Know temperature exactly, pressure uncertain (known variance).
Conditioning on Y: Know both exactly, no remaining uncertainty.

11. $X = \pi_1 \circ Y$

$\pi_1(t, p) = t$ is continuous, hence measurable; Y identity, measurable; composition measurable $\Rightarrow X$ measurable w.r.t. $\sigma(Y)$.

12. Differential entropy of X

$$h(X) = \frac{1}{2} \ln(2\pi e) \approx 1.4189 \text{ nats} \approx 2.047 \text{ bits}.$$

13. Differential entropy of Y

By independence,

$$h(Y) = h(T) + h(P) = \frac{1}{2} \ln(2\pi e) + \frac{1}{2} \ln(8\pi e) = \frac{1}{2} \ln(16\pi^2 e^2) \approx 3.5310 \text{ nats} \approx 5.093 \text{ bits}.$$

14. $h(Y|X)$

Given $X = t$, P remains $N(0, 4)$, so

$$h(Y|X) = h(P) = \frac{1}{2} \ln(8\pi e) \approx 2.1121 \text{ nats } (\approx 3.046 \text{ bits}).$$

Physically: uncertainty in Y after knowing X is just uncertainty in P.

15. $h(X|Y)$

Since X is a deterministic function of Y, in the continuous entropy framework:

$$h(X|Y) = 0.$$

Difference from discrete: here "zero" means no differential entropy left, matching intuition that knowing Y removes all uncertainty in X.